

PRST Clam (10/11/2020)

X is said to follow exponential dist with parameter λ ($\text{Exp}(\lambda)$)

if $f(x) = \lambda e^{-\lambda x}, x > 0, \lambda > 0$

cf $F(x) = 1 - e^{-\lambda x}$

$$E(X) = \int_0^{\infty} x \lambda e^{-\lambda x} dx$$

$$= \int_0^{\infty} \frac{t}{\lambda} e^{-t} dt$$

$$= \frac{\Gamma 2}{\lambda}$$

$$= \frac{1}{\lambda}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx$$

$$= \int_0^{\infty} x \left(\frac{t}{\lambda}\right)^2 e^{-t} \frac{dt}{\lambda}$$

$$= \frac{1}{\lambda^2} \int_0^{\infty} t^2 e^{-t} dt$$

$$= \frac{\Gamma 3}{\lambda^2} = \frac{2}{\lambda^2}$$

$$\lambda x = t$$

$$dx = \frac{1}{\lambda} dt$$

$$\Gamma a = \int_0^{\infty} x^{a-1} e^{-x} dx$$

$$\lambda x = t$$

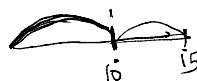
$$dx = \frac{dt}{\lambda}$$

$$V(X) = \frac{1}{\lambda^2}$$

Lack of Memory Property:

$$P(X > a+b | X > a) = P(X > b)$$

$$a = 10, b = 5$$



$$P(X > a+b | X > a) = \frac{P(X > a+b, X > a)}{P(X > a)}$$

$$= \frac{P(X > a+b)}{P(X > a)}$$

$$= \frac{e^{-\lambda(a+b)}}{e^{-\lambda a}} = e^{-\lambda b}$$

$$\begin{aligned} F(x) &= 1 - e^{-\lambda x} \\ P(X \leq x) &= 1 - e^{-\lambda x} \\ P(X \geq x) &= e^{-\lambda x} \end{aligned}$$

$$= \frac{P(\lambda(a+b))}{\frac{e^{-\lambda a}}{e^{-\lambda a}}} = \frac{e^{-\lambda(a+b)}}{e^{-\lambda a}} = e^{-\lambda b}$$

$$P(X > b) = e^{-\lambda b}$$

$$= P(X > b)$$

for Geometric dist: $G(p)$

$$P(X \geq s+t | X \geq t) = P(X \geq s)$$

$$P(X \geq n) = q^n$$

$$\begin{aligned} \sum_{t=n}^{\infty} P(X \geq t) &= \sum_{t=n}^{\infty} P q^{t-1} \\ &= P \left(q^n + q^{n+1} + \dots \right) \\ &= P q^n (1 + q + q^2 + \dots) \\ &= P q^n \frac{1}{1-q} \\ &= q^n \end{aligned}$$

$$P(X \geq s+t | X \geq t)$$

$$= \frac{q^{s+t}}{q^t} = q^s = P(X \geq s)$$

$\mathcal{L}(0,1)$ - standard Cauchy

Cauchy - $f(x) = \frac{1}{\pi} \frac{1}{1+x^2}$; $-\infty < x < \infty$

$$Z = \frac{X-u}{\lambda} \quad X \sim \mathcal{L}(u, \lambda)$$

$$f(x) = \frac{1}{\pi} \cdot \frac{1}{\lambda} \frac{1}{1 + \left(\frac{x-u}{\lambda}\right)^2}$$

$$= \frac{1}{\pi} \cdot \frac{1}{\lambda} \frac{\lambda^2}{\lambda^2 + (x-u)^2}$$

$$= \frac{1}{\pi} \frac{\lambda}{\lambda^2 + (x-u)^2}$$

$$\begin{array}{l} X \sim f(x) \\ Y = g(X) \end{array}$$

Ex. The time a customer waiting in a bank is exponential dist with mean 10 m.
What is the prob that the customer will wait even after 5 m. (5 h)?

Sol:

$$X \sim \text{Exp}(\lambda)$$

$$E(X) = \frac{1}{\lambda} = 10$$

$$\Rightarrow \lambda = \frac{1}{10}$$

$$P(X > 5) = e^{-\lambda \cdot 5} = e^{-\frac{5}{10}} = e^{-\frac{1}{2}}$$

$\lambda e^{-\lambda x}$	$\lambda = \frac{1}{\theta}$
$E(X) = \frac{1}{\lambda}$	$\frac{1}{\theta} = \frac{x}{\theta}$
	$\theta = E(X)$

Ex. Time spent on a computer X is gamma dist with mean 20, var 80.
What are values of α and β .

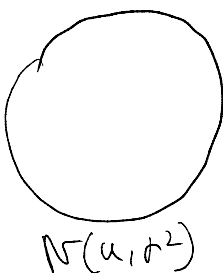
$$E(X) = \frac{\beta}{\alpha} = 20$$

$$\text{Var} = \frac{\beta}{\alpha^2} = 80$$

$$\frac{\alpha^\beta}{\Gamma(\beta)} x^{\beta-1} e^{-\alpha x}$$

Sampling distribution

μ, σ



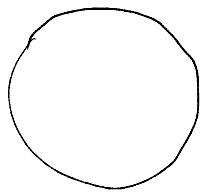
F_0

F_0

$$\theta = (\mu, \sigma^2)$$

... is characterised

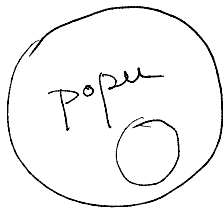
$$N(\mu, \sigma^2) \quad \text{TV}$$



The population is characterised by the dist $F_\theta(\cdot)$ $\left| \begin{array}{l} F(\cdot, \theta) \\ F(\cdot | \theta) \end{array} \right.$

$\theta \rightarrow$ scalar or vector

$$\theta \in \mathbb{R}, \mathbb{R}, \dots \mathbb{R}^d$$



Sample with replacement
without replacement.

$a \times b$

X_1 - rv denotes the gene profile of the 1st person in the sample



X_{100} - a 100th person

X_i - rv a ith person

$n \quad (X_1, X_2, \dots, X_n)$ - simple random sample.

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{sample mean}$$

$$\frac{1}{n} \sum (X_i - \bar{X})^2 \quad \text{sample variance.}$$

Statistic - Any function of random sample.